

Exercise 7

Implementing Classes 2: Cylindrical Can Drain and Fill Problem

Introduction:

When the code base of a problem will go in size so will its complexity. But when we think of the entire program as a single entity, composed of autonomous sub-entities, then we can simplify its complexity. In computer science, abstracting information for the perusal of the user is the first step in describing the features of a program. The class in C++ is the means in which the abstraction can be translated into a user-defined type.

For this exercise, the cylindrical can would be represented as an object. The volume is the property used to describe the cylindrical can, and it determines the total capacity of the liquid it may hold. The equation is as follows, $v = \pi * r^2 * h$

Learning Outcomes:

- Create a class for a cylindrical can with the appropriate member variables and functions.
- Implement and test the functions that are in the class by creating an instance of the class (object).

Problem Background

1. Create a project file for this exercise.
2. On a header file, cylindricalCan.h, create a class named cylindricalCan with the private variables indicated as follows:

```
double height;           // the height of the can (meters)
double radius;           // the radius of the can (meters)
double currentVolume;    // current liquid volume (liters)
double drainRate;        // drain rate of the can (liters/sec)
double fillRate;         // fill rate of the can (liters/sec)
```

3. In your class, write the function prototypes as indicated below:

```
cylindricalCan();
    // default constructor, sets all variables to 0
void setHeight(double);           // set the height of the can
void setRadius(double);          // set the radius of the can
void setCurrentVolume(double);
    // set the current liquid volume
void setDrainRate(double);
    // set the drain rate of the can
void setFillRate(double);        // set the fill rate of the can
double getHeight();
```

```

        // return the value of the height
double getRadius();
        // return the value of the radius
double getCurrentVolume();
        // return the current liquid volume
double getDrainRate();
        // return the value of the drain rate
double getFillRate();
        // return the value of the fill rate
double totalVolume();
        // computes and returns the total capacity
~cylindricalCan();
        // default destructor, no content on function body

void secondsToDrain();
/* computes and displays the time in seconds to fully drain
   the liquid from the current volume at the drain rate. */

void secondsToFill();
/* computes and displays the time in seconds to fully fill
   the can from the current volume at the fill rate. */

void secondsToFillPartial(double);
/* computes and displays the time in seconds to fill the
   can from the passing parameter volume to total capacity. */

```

4. On another file cylindricalCanImp.cpp, implement the function body for the class cylindricalCan.

5. Test the operation of these functions on main.cpp function main().